Integrating a One-Zone Pulsar Wind Nebula Model with Gammapy

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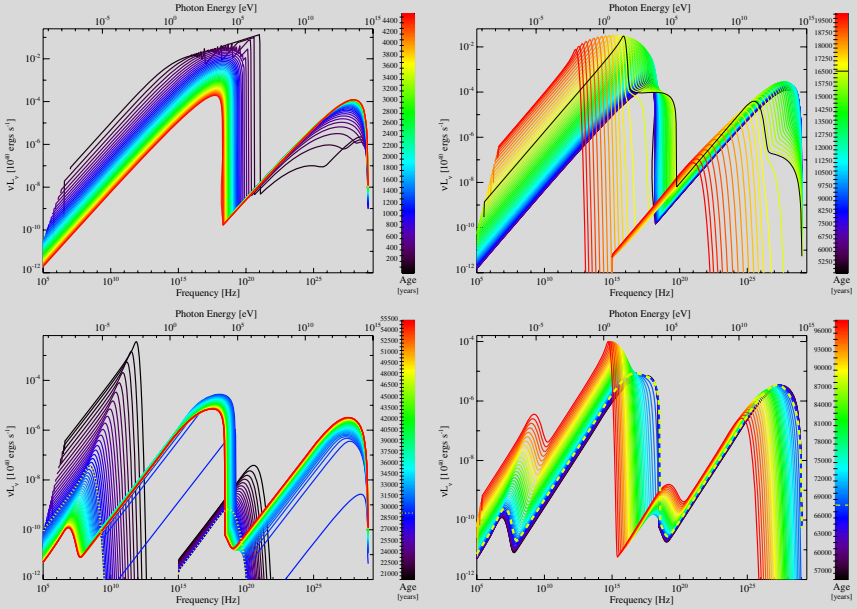

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Motivation & Context

- PWNe are the most common class of Galactic TeV source — LHAASO 1st catalog: 43 UHE sources, many PWN/halo candidates
- Modeling a PWN requires coupling dynamics (radius, pressure, B-field) to radiation (radio → TeV) self-consistently over the full SNR lifetime
- Existing tools (current gammapy built-in models) describe static particle populations, but they don't evolve the system forward in time
- pwnmodel fills this gap: given SN + pulsar + ISM properties, it predicts the broadband SED, PWN size, SNR size and dynamics at any age
- Fit physical parameters directly to multi-wavelength data, not just a spectral shape

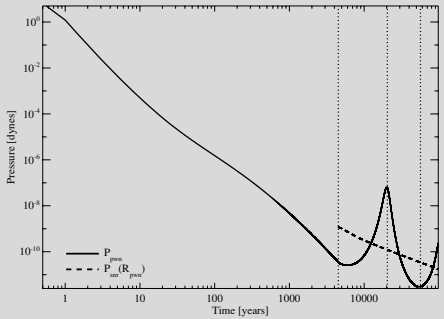


Figure 3. Pressure inside the PWN ($P_{\rm pwn}$, solid) and the pressure inside the SNR just outside the PWN ($P_{\rm snr}(R_{\rm pwn})$, dashed). At times $t < t_{\rm col}$, we assume that $P_{\rm snr}(R_{\rm pwn}) = 0$. The vertical dotted lines indicate the division between the four evolutionary phases of the PWN discussed in Section 3.

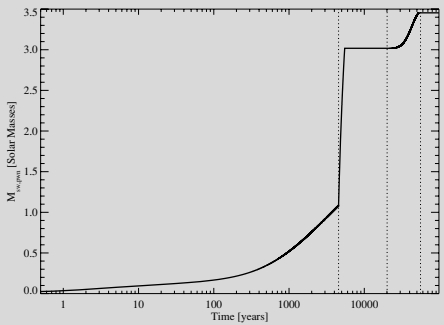
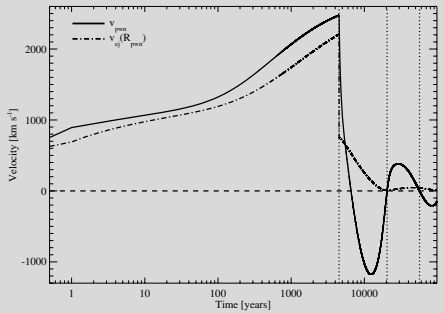
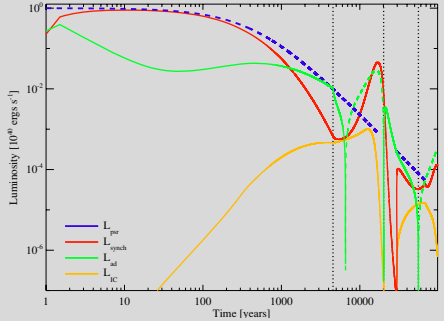


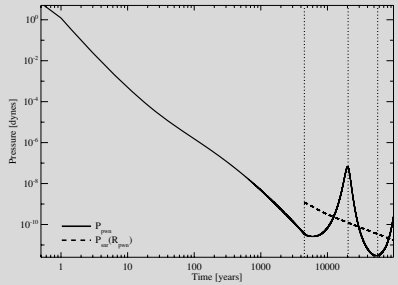
Figure 5. Mass of material swept up by the PWN ($M_{\rm sw,pwn}$) as it expands inside the SNR. The vertical dotted lines indicate the division between the four evolutionary phases of the PWN discussed in Section 3.





Model Physics: Dynamics

- Based on Gelfand, Slane & Zhang (2009, ApJ 703, 2051)
- SNR dynamics: Truelove & McKee (1999) analytic solution for non-radiative SNR, n=9 ejecta profile
- PWN modeled as a thin shell of swept-up ejecta; radius driven by $\Delta P \times 4\pi R^2$
- Adaptive timesteps: Courant-like conditions on radius, velocity, cooling time, RS proximity
- Bandiera (1984) self-similar pressure/density profile after reverse-shock collision
- Key phases: free expansion → RS collision → compression → re-expansion



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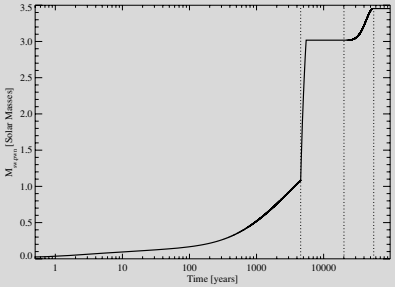

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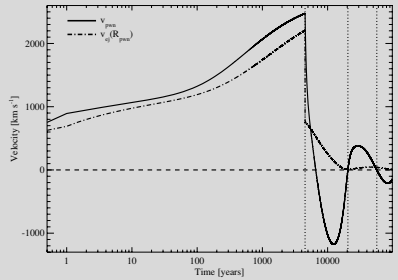
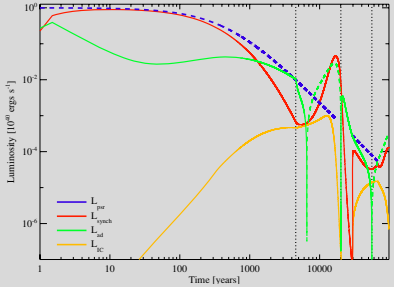
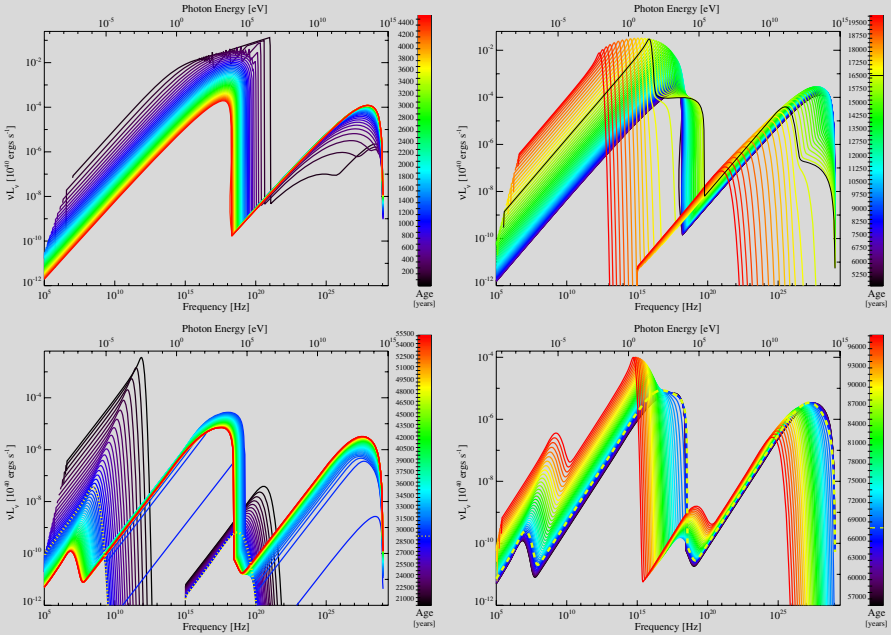




Model Physics: Radiation

- $B \propto R_pwn^{-2}$ (flux conservation) + continuous injection at magnetization η_B
- Particle injection: broken power-law (p1, p2, E_break) + optional relativistic Maxwellian
- 500-bin electron spectrum evolved each timestep: adiabatic + synchrotron + IC cooling
- Synchrotron: full F(x) = x∫K_{5/3} Bessel function (not monochromatic approx.)
- IC: exact Klein-Nishina cross-section on CMB + additional thermal photon fields (200 target bins)
- Broadband SED output: radio through >100 TeV



Model Parameters

Category	Parameters
Supernova	E_SN (10^{51} erg), M_ej ($M_\odot$)
ISM	n_ISM (cm^{-3})
Pulsar	P, $\dot{P} \rightarrow \dot{E}_0$, τ_sd, age; braking index, η_g, v_psr
Wind	η_B, E_min, E_max, E_break (GeV), p1, p2
Maxwellian	f_max (fraction), kT_max (GeV)
IC fields	T (K), norm for each thermal field beyond CMB
Distance	d (pc) — converts luminosities to fluxes

- Typical fit: 12–16 free parameters
- Outputs: R_pwn, R_snr, B-field, energetics vs time; electron spectrum N(E,t); photon SED L_ν(ν,t)

```
Parameter           delta      value          bounds
--------------------------------------------------------------------------------
esn               ↑1.059x      1.063 e51 erg  [0.01, 30]
mej               ↑2.243x      13.37 Msun     [0.5, 100]
n_ism             ↑1.137x    0.06815 cm-3     [0.001, 10]
braking_index     ↑1.109x      2.957          [1.5, 3.5]
tau_sd            ↑1.480x       4710 yr       [100, 1e+05]
eta_b             ↓0.336x    0.09841          [0.0001, 0.5]
e_min             ↑1.141x      1.143 GeV      [0.1, 1e+03]
e_max             ↑1.502x      379.3 TeV      [10, 1e+04]
e_break           ↑26.263x       2622 GeV      [1, 1e+05]
p1                ↓0.481x     0.7227          [0.5, 2.5]
p2                ↑1.071x       2.68          [1.5, 3.5]
distance          ↓0.837x      1.453 kpc      [0.4, 3]  prior(1400.0,400.0) chi2=0.02

Dynamics: R_pwn=6.14 pc, R_snr=22.32 pc, B=3.215 uG
  Age = 9489 yr (9.5 kyr), v_pwn = 729 km/s
  Reverse shock has not reached PWN

Observable                          model       target    pull   wt  wt*chi2
--------------------------------------------------------------------------------
θ_snr [arcmin]                       52.8           45   +0.78  1.0     0.61
θ_pwn [arcmin] (10.2-13.8)           14.5  [10.2,13.8]   +0.40  1.0     0.16
VLA 1.4 GHz [UL]                 8.83e-17    <1.24e-15      OK  1.0     0.00
CXO 0.5-7.0 keV [LL]             8.13e-12     >1.4e-12      OK  1.0     0.00
ASCA 0.5-7.0 keV [UL]            8.13e-12     <8.4e-12      OK  1.0     0.00
Fermi 67 GeV                     3.35e-12      3.5e-12   -0.17  1.0     0.03
Fermi 123 GeV                    3.58e-12     2.69e-12   +0.79  1.0     0.63
Fermi 224 GeV                    3.72e-12     1.75e-12   +1.19  1.0     1.41
Fermi 407 GeV                    3.82e-12     3.02e-12   +0.46  1.0     0.22
Fermi 741 GeV                    3.84e-12     5.89e-12   -0.59  1.0     0.35
WCDA 7.5 TeV                     3.01e-12     3.46e-12   -0.79  1.0     0.62
WCDA 11.8 TeV                    2.65e-12     2.95e-12   -0.65  1.0     0.43
WCDA 17.2 TeV                    2.32e-12     2.44e-12   -0.31  1.0     0.10
WCDA 30.5 TeV                    1.77e-12     2.15e-12   -0.99  1.0     0.98
WCDA 57.3 TeV                    1.17e-12     1.23e-12   -0.19  1.0     0.04
KM2A 49 TeV                      1.32e-12     1.27e-12   +0.25  1.0     0.06
KM2A 79 TeV                      8.78e-13     9.72e-13   -0.94  1.0     0.89
KM2A 109 TeV                     6.13e-13     5.67e-13   +0.63  1.0     0.40
KM2A 175 TeV                     2.83e-13     2.78e-13   +0.08  1.0     0.01
KM2A 250 TeV                     9.89e-14     1.09e-13   -0.21  1.0     0.05

  Unweighted obs chi2 = 6.96
  Reduced chi2        = 6.96/8 = 0.87
```

Python Implementation

- Original: ~3100-line C++ code (pwnmodel_gold.cc, 2009– present) → Python/NumPy rewrite (v0.3.0)
- pip install -e ".[all]" — Python ≥3.10, NumPy, SciPy, Astropy; optional Numba (10× speedup), emcee, gammapy
- 13 modules: model.py, evolution.py, radiation.py, particles.py, snr.py, observables.py, fitting.py, catalog.py, gammapy_model.py, …
- Vectorized evolution: eliminates 21.5M Python function calls per run
- Fused IC kernel: eliminates 1 GB intermediate 3D array; optional Numba JIT (0.1–0.3 s vs 2.5 s fallback)
- Cached F(x) synchrotron lookup table (2000 pts, built once at import)
- Result: ~1.4 s per evaluation on Apple M3 Max with Numba
- Validated against C++ reference
- Emcee integration complete.

Installation

Requirements: Python >= 3.10, NumPy >= 1.21, SciPy >= 1.7, Astropy >= 5.0

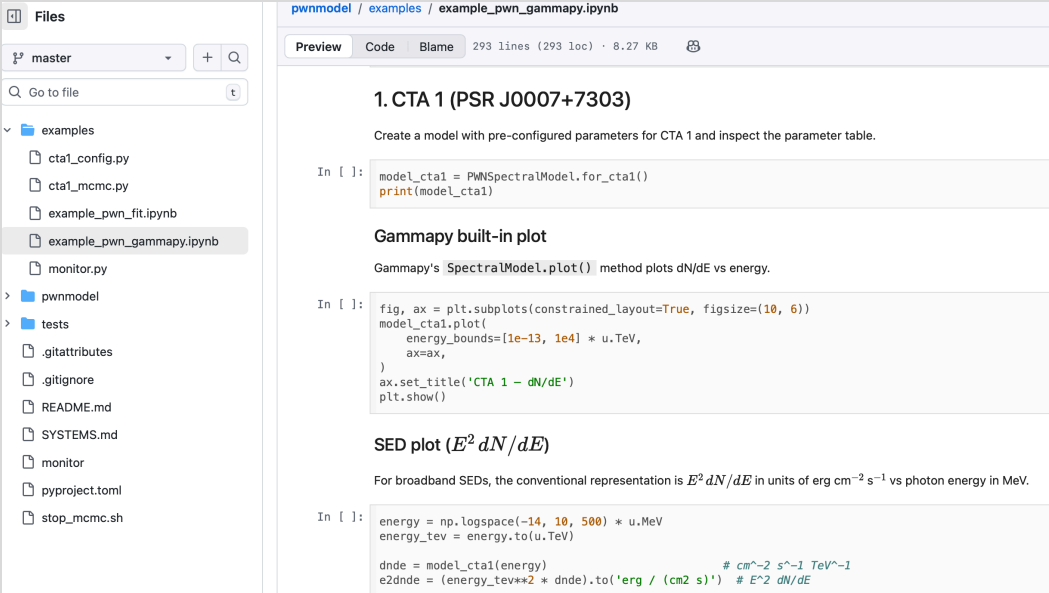
```
git clone https://github.com/j-alford/pwnmodel.git
cd pwnmodel
pip install -e ".[fast]"      # recommended: includes Numba for 10x speedup
```

Quick Start

```
import astropy.units as u
from pwnmodel import PWNModel

model = PWNModel()
model.set_pulsar('J0007+7303')           # loads P, Pdot, distance from ATNF catalog

model.esn.value = 1e51 * u.erg
model.mej.value = 6.0 * u.M_sun
model.n_ism.value = 0.4 / u.cm**3
model.braking_index.value = 2.5
model.tau_sd.value = 3000 * u.yr
model.eta_b.value = 0.01
model.e_min.value = 3.0 * u.GeV
model.e_max.value = 500 * u.TeV
model.e_break.value = 50.0 * u.GeV
model.p1.value = 0.8
model.p2.value = 2.5

result = model.run()

dyn = result.dyninfo[-1]
print(f"R_pwn = {dyn.rad_pwn:.2f} pc")
print(f"B     = {dyn.bpwn:.2f} uG")
print(f"V_pwn = {dyn.vpwn:.0f} km/s")
```

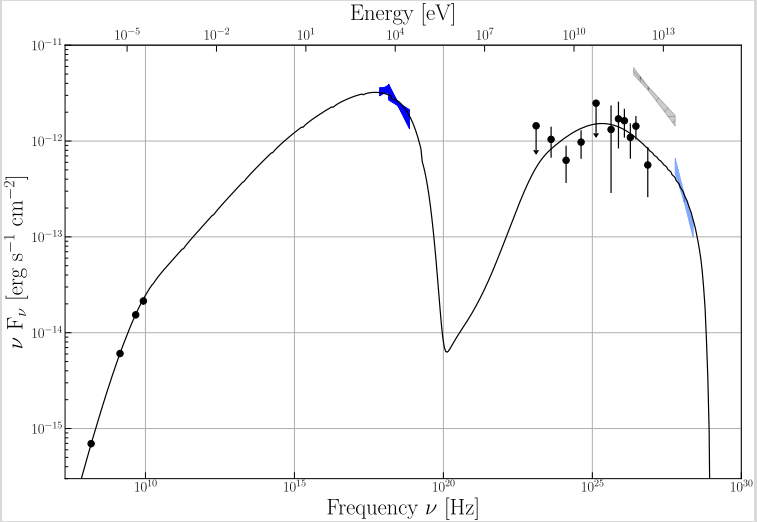
Gammapy Integration

- PWNSpectralModel — a gammapy SpectralModel subclass
- All PWN physics params exposed as gammapy Parameter objects: E_SN, M_ej, n_ISM, η_B, τ_sd, p1, p2, E_min, E_max, E_break, …
- evaluate() runs the full time-dependent evolution, returns dN/dE [cm^{-2} s^{-1} TeV^{-1}]
- Two-layer caching: parameter hash skips ~1.4 s model run; log-linear interpolation onto any energy grid (μs)
- Pre-configured: PWNSpectralModel.for_cta1(), .for_g54(), more coming soon!
- Up to 2 additional thermal IC fields (unfreeze to fit)
- Plugs directly into gammapy Fit framework with any Dataset
- Also works standalone without gammapy



Recent Publications

System	Pulsar	Reference
Crab Nebula	B0531+21	Gelfand+2009
CTA 1	J0007+7303	Aliu+2013
Kes 75	J1846-0258	Gotthelf+2021
G54.1+0.3	J1930+1852	Gelfand+2015, Alford+2025
HESS J1640-465	J1640-4631	Abdelmaguid+2023,2025
Boomerang	J2229+6114	Pope+2024
Dragonfly	J2021+3651	Woo+2023
1LHAASO J1740+0948u	J1740+1000	Brunelli+2026
+ G21.5, Eel, MSH 15-56, G327.1, B0453-685		



Summary & Future Work

- Physics-driven PWN spectral model → validated Python package → gammapy integration
- Fit physical parameters (E_{SN}, η_B, τ_{sd}, …) directly to γ-ray data, not just spectral shapes
- 13 systems modeled across 20+ publications
- Package: pip install pwnmodel/github.com/j-alford/pwnmodel
- Future: IC lookup-table library for faster MCMC run, Numba acceleration of remaining hot paths, SSC support
- Potential rename? ‘pwnmodel’ is generic…